



HARSHAL V. TARAR

Mechanical Engineer with 10 years of experience across NPD, manufacturing scale-up, APQP/PPAP, and Quality Systems Management in automotive, electronics, and advanced materials.

Founder of IKKA, focused on building governed AI decision infrastructure that turns fragmented enterprise knowledge into traceable, auditable, and reusable engineering decisions.

[GitHub](#) | [LinkedIn](#) | harshal@ikkasolutions.com

EDUCATION

- M.S., Mechanical Engineering (Design & Manufacturing) Sep 2014 – May 2015
University of Massachusetts Lowell, Lowell, MA, USA
- B.S., Mechanical Engineering Sep 2011 – May 2014
University of Massachusetts Lowell, Lowell, MA, USA

WORK EXPERIENCE

Founder Sep 2025 – Present
[IKKA Solutions](#) (<1 Year)
Thane, MH, India

- Founded IKKA (meaning “Ace”) as an independent platform for engineering systems, quality operations, and manufacturing intelligence, focused on helping teams identify recurring gaps where fragmented documentation, tacit know-how, and cross-functional ambiguity create rework, decision lag, and audit risk.
- Deconstructed APQP, PPAP, CSRs, and Launch Readiness workflows into structured decision maps and early knowledge-system scaffolding; revealing an opportunity to convert fragmented enterprise knowledge and engineering rationale into traceable, auditable, and reusable decision artifacts.
- Evolved IKKA into a pre-seed, product-driven initiative by building a working MVP with a proprietary API layer and Custom GPT interface, using deterministic context, evidence retrieval, controlled synthesis, and decision traces as the foundation for governed AI infrastructure; current focus is technical co-founder alignment, product hardening, and pilot validation.

Process Quality Engineer Jan 2022 – Dec 2024
[Aspen Aerogels Inc.](#) (<3 Years)
Marlborough, MA, USA

- Served as primary customer-facing Quality lead across Automotive Thermal Barrier programs supporting GM, Toyota, Porsche-Valmet, ACC, Mercedes-Benz and Stellantis; contributed to execution during Aspen’s reported Thermal Barrier revenue ramp from ~\$7M in 2021 revenue to ~\$310M in 2024.
- Led global line qualification, PPAP readiness, vendor qualification, inspection strategy, and customer validation across internal, Mexico, and UK contractor sites; supporting transition from pilot builds to high-volume production and customer-facing launch readiness.
- Delivered audit and qualification leadership across IATF, VDA 6.3, and CSR-driven requirements, including Scania / Volkswagen Group supplier approval activities that enabled broader VW Group business participation.

- Led automation and systems improvements, including ~\$1M QC CAPEX, cycle-time reduction from ~95s to ~5.5s per part, and AI-integrated automated inspection concepts with estimated YoY savings of ~\$430k.

Sr. Quality Engineer

Jun 2021 – Nov 2021

[SharkNinja](#)

(<1 Year)

Needham, MA, USA

- Led quality assurance and product development strategies for key kitchen appliance categories, integrating Voice of Customer (VoC) insights into New Product Development (NPD) cycles to maximize consumer satisfaction.
- Developed KPI analytics for weekly executive reviews, while also providing operational leadership for the Failure Analysis laboratory to inform and drive product quality enhancements.

Process Design Engineer II

Sep 2016 – Jun 2021

[Sensata Technologies Inc.](#)

(<5 Years)

Attleboro, MA, USA

- Led the development of MEMS and Strain Gage (MSG) based automotive sensor applications, translating design and customer requirements into manufacturing process flows and driving equipment selection, installation, and qualification.
- Directed process development from functional prototype to mass production across manufacturing technologies such as Resistance and Laser Welding, PCB Calibration, Profile Measurement, Computer Vision, Function Testing, Wire Bonding and Dispense, including managing over \$1M in CAPEX
- Managed critical aspects of the global supply chain, developing supplier qualification plans, leading root cause investigating of quality events and supporting dual-sourcing strategies.

Quality Co-Ordinator

Oct 2015 – Aug 2016

[Dow Chemical Co.](#) (now Qnity Electronics Inc.)

(<1 Year)

Marlborough, MA, USA

- Site Quality Leader for one of two business units, directly resolving customer inquiries on product conformance from market-driving customers like Samsung, GlobalFoundries and BASF
- Primary on-site stakeholder to investigate non-conforming products via Statistical Process Control (SPC), raw material mapping, and lead cross-functional investigations.
- Led the upgrade of a Manufacturing Execution System (MES) database to automate data tracking, reduce operator error, and double critical to quality (CTQ) parameters tracking, resulting in improved customer outcomes.

Manufacturing Engineer Co-Op

Jan 2013 – May 2014

[Vicor Corporation](#)

(<1 Year)

Andover, MA, USA

- Optimized robotic machinery programs by rectifying GD&T and pattern recognition errors to reduce process interruptions and increase cycle time.
- Designed and implemented a process for Silicone based Thermal Interface Material (TIM) application in the PCB over-molding processes in order to achieve consistent ferrite core gaps and improve functional capability.
- Led Kaizen initiative to evaluate effectivity and investment potential of new water wash process by performing comparative investigation of solder flux and voiding incidence.

Sales Associate	Aug 2013 – May 2014
Follett Corp. – UMASS Lowell Campus Bookstore	(<1 Year)
Engage customers to assist with shopping needs and provide administrative assistance	

ACADEMIC EXPERIENCE

Research Lab Assistant	Oct 2014 – Sep 2015
<u>Advanced Composite Materials and Textiles Research Lab</u>	(<1 Year)
UMASS Lowell, Lowell, MA, USA	
- Characterized properties of flame-retardant, vinyl ester resins for structural composites - Designed and fabricated digital composite materials capable of in-situ structural health monitoring - Developed manufacturing methods and ASTM test methods for self-healing composites	
Teaching Assistant	Oct 2014 – May 2015
<u>Department of Mechanical Engineering</u>	(<1 Year)
UMASS Lowell, Lowell, MA, USA	
Supervised and guided the Instrumentation and Measurement Laboratory and Thermal Fluids Laboratory courses for junior and senior-level undergraduate courses.	

SKILLS

- Manufacturing Process Development – DFSS, FMEA, DOE, MSA, URS, FAT, CT Improvements
- Quality Management – APQP, CAPA, QMS Development, Internal Audits (ISO, IATF, VDA)
- Data Analysis & Visualization – Minitab, JMP, Excel, SQL, Power BI, Visio
- Engineering & Design – SolidWorks, Abaqus, Moldflow, MATLAB
- Enterprise Systems – Agile, JIRA, SAP

CERTIFICATIONS

- Engineering Fundamentals & Design:
 - Engineer In Training (EIT) – Jan 2015
 - Certified SolidWorks Associate – Jun 2015
 - Geometric Dimensioning and Tolerancing (GD&T) – Nov 2017
- Quality Management & Auditing:
 - VDA 6.3 Process Audit – Feb 2022
 - Internal Auditor (ISO 9001:2015 & IATF 16949:2016) – Dec 2022
 - AIAG Core Tools Workshop – May 2017
 - Six Sigma Methodologies (Critical Thinking) – Oct 2017
- Data Science & Business Analytics:
 - Database Management System using MySQL – Oct 2025
 - Data Analytics with Power BI – Oct 2025

VOLUNTEERING

- STEM Mentor – Guided middle and high school students in STEM fundamentals through engaging, thought-provoking experiments, fostering a fun and interactive learning environment.
- United Way of America – Participated in community initiatives to restore of a 100-foot boardwalk for the Attleboro Land Trust Nature Preserve, enhancing local recreational access.
- International Student Ambassador – Planned, budgeted, and executed a range of socio-cultural, academic, and official events to support and integrate international students within the university community.

HONORS & AWARDS

- Emerald Award – Recognized at Dow Chemical for outstanding efforts in driving internal quality improvements that consistently met customer expectations.
- Award of Recognition – Recognized at Vicor Corporation for exceptional performance and exceeding expectations in manufacturing engineering.
- International Academic Council Scholarship (Fall 2011 to Spring 2013) – Awarded in recognition of sustained academic excellence for Undergraduate studies.
- Dean's List (Fall 2011, Fall 2013) – Awarded in recognition of a semester GPA 3.25 – 3.50 during Undergraduate studies.

HOBBIES & INTERESTS

- Sport – Tennis, Golf
- Leisure – Cooking, Painting
- Learning – Art History, History
- Professional Growth – Agentic Workflow Development, Net Zero Technologies